

## 38 – Environment TagHelper

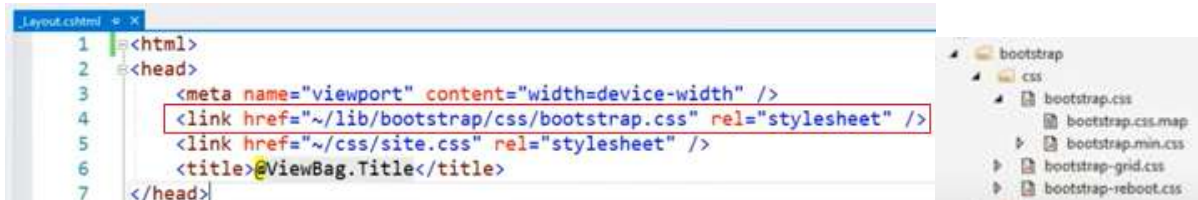
1) Искаме на локалната машина (our own server) при \* Development -> bootstrap.css

2) В останалите случаи при \* Staging

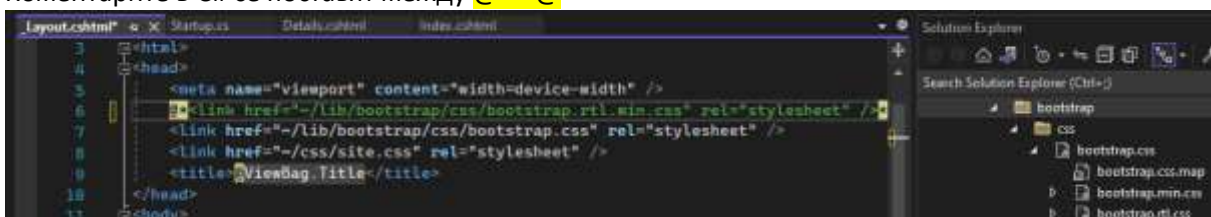
\* Production -> bootstrap.min.css

\* UAT

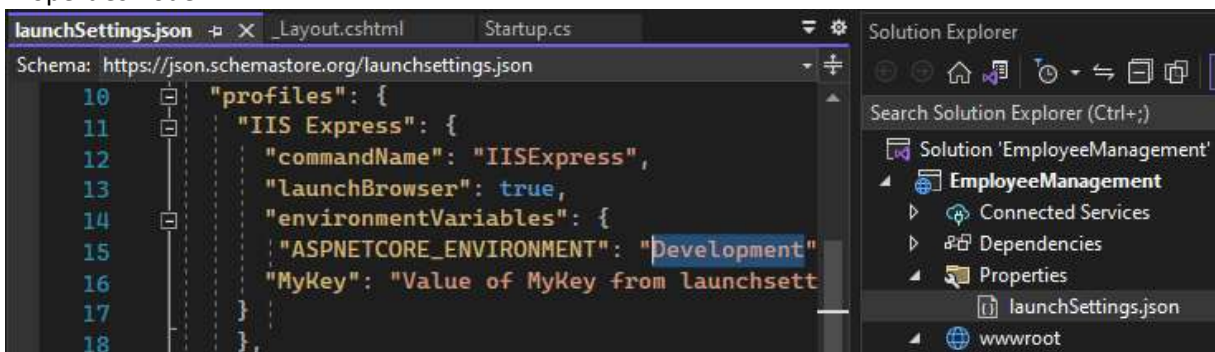
и ако CDN (Content Delivery Network) is down -> bootstrap.min.css



Коментарите в C# се поставят между @\* \*@



Properties node:



Staging/Production in OS.

# ASP.NET Core Environment Tag Helper

- The application environment name is set using **ASPNETCORE\_ENVIRONMENT** variable

```
<environment include="Development">
  <link rel="stylesheet"
        href="/lib/bootstrap/css/bootstrap.css" />
</environment>
```

- Environment tag helper supports rendering different content depending on the application environment

For the development environment render this content that is enclosed within the <environment> tag

```
<environment include="Development">
  <link href="/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
</environment>
```

```
<environment include="Development">
  <link href="/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
</environment>
```

## include and exclude attributes

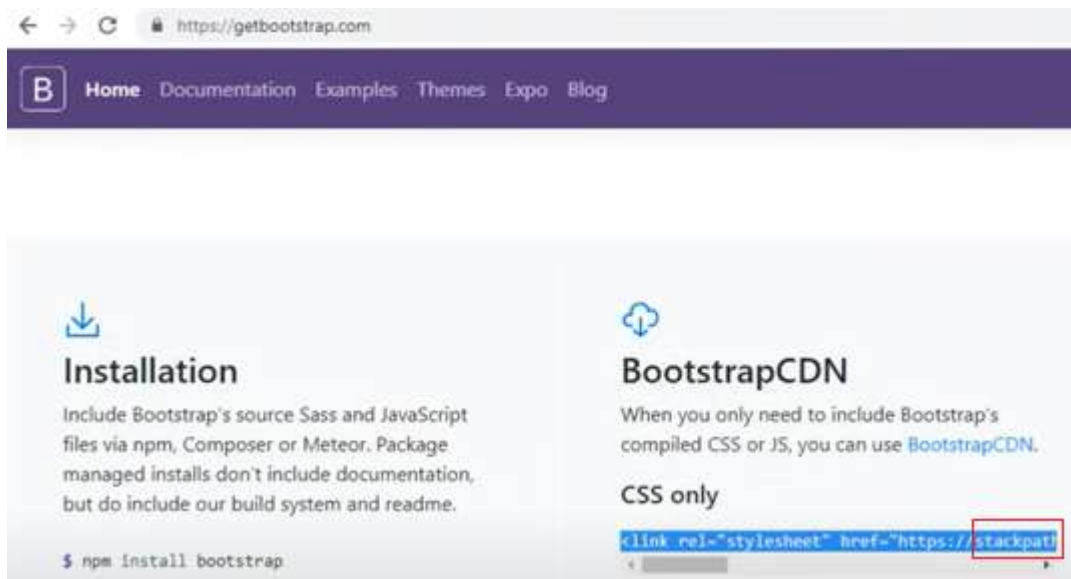
```
I <environment include="Staging,Production">
    <link rel="stylesheet"
          href="https://CDN_URL/css/bootstrap.min.css"
          integrity="sha384-Integrity_Hash....">
</environment>
or
II <environment exclude="Development">
    <link rel="stylesheet"
          href="https://CDN_URL/css/bootstrap.min.css"
          integrity="sha384-Integrity_Hash....">
</environment>
```

The value of **"include"** and **"exclude"** attribute can be either a single hosting environment name or a comma-separated list of hosting environment names

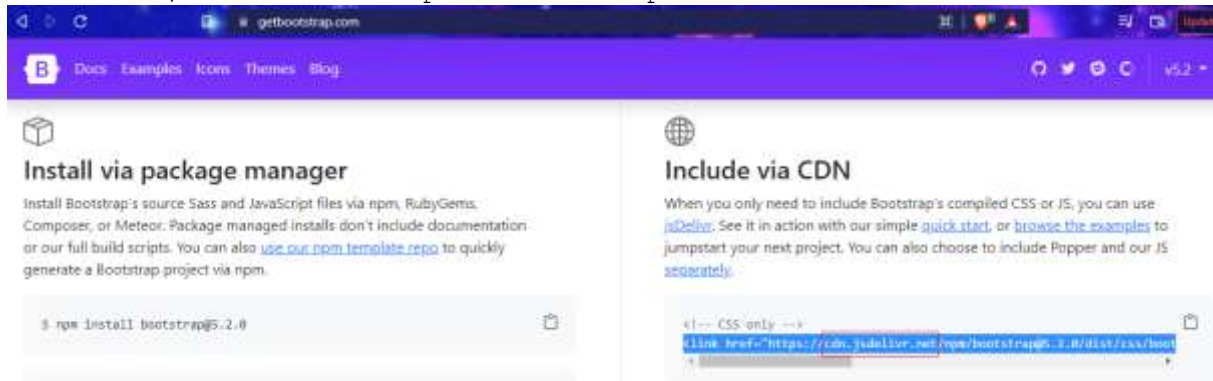
Include - For Staging or Production use minified css bootstrap file from CDN URL link (изброяваме, разделени със запетайки).

Exclude - „с изключение на“ е удобен, когато впоследствие се наложи да добавим нов environment like UAT, няма да се налага да правим никаква промяна. При include ще трябва да добавим новата UAT разделена със запетая от останалите.

<https://getbootstrap.com/>

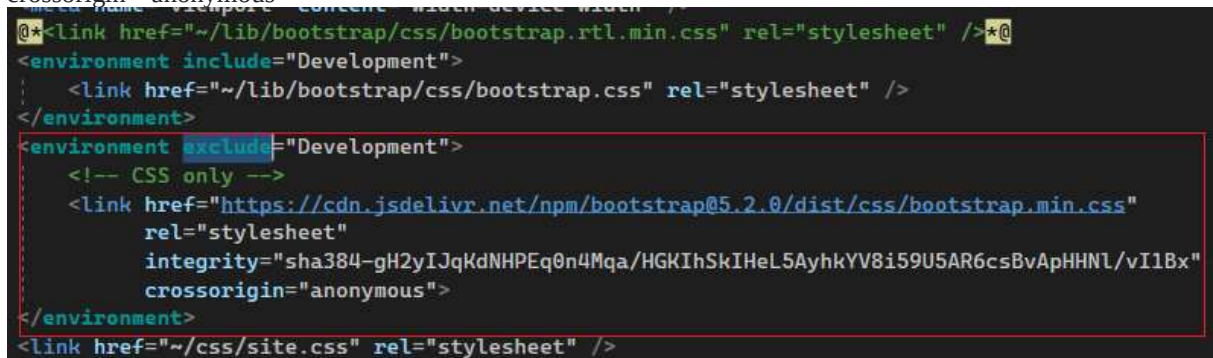


Забелязвам, че в новата версия линкът е различен:



<!-- CSS only -->

```
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css" rel="stylesheet"
integrity="sha384-gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5AR6csBvApHHNl/vI1Bx"
crossorigin="anonymous">
```



```
<environment exclude="Development">
```

```
<!-- CSS only -->
```

```
<link
```

```
href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
rel="stylesheet"
integrity="sha384-
gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5AR6csBvApHHNl/vI1Bx"
crossorigin="anonymous">
```



</environment>

```
view-source:localhost:24077
<meta name="viewport" content="width=device-width" />
<link href="/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
    our own server/local machine
<link href="/css/site.css" rel="stylesheet" />
<title>Employee List</title>
```

Нека сменим на Staging:

```
launchSettings.json  _Layout.cshtml  Startup.cs  Details.cshtml
Schema: https://json.schemastore.org/launchsettings.json
10  "profiles": {
11    "IIS Express": {
12      "commandName": "IISExpress",
13      "launchBrowser": true,
14      "environmentVariables": {
15        "ASPNETCORE_ENVIRONMENT": "Staging",
```

Веднага се зарежда bootstrap.min.css

```
view-source:localhost:24077
<meta name="viewport" content="width=device-width" />
<!-- CSS only -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
rel="stylesheet"
integrity="sha384-gH2yI1qKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkyV8i59U5AR6csBvAphHhNl/vI18x"
crossorigin="anonymous">
<link href="/css/site.css" rel="stylesheet" />
```

integrity – SubResource integrity (SRI) check - проверка за сигурност дали файлът е изтегнен от сървъра или има злонамерено вмешателство.

## SubResource Integrity (SRI)

```
<environment exclude="Development">
  <link rel="stylesheet"
    href="https://CDN URL/css/bootstrap.min.css"
    integrity="sha384-Integrity_Hash....">
</environment>
```

- The **"integrity"** attribute on the **<link>** element is used for Subresource Integrity check
- SRI is a security feature that allows a browser to check if the file being retrieved **has been maliciously altered**

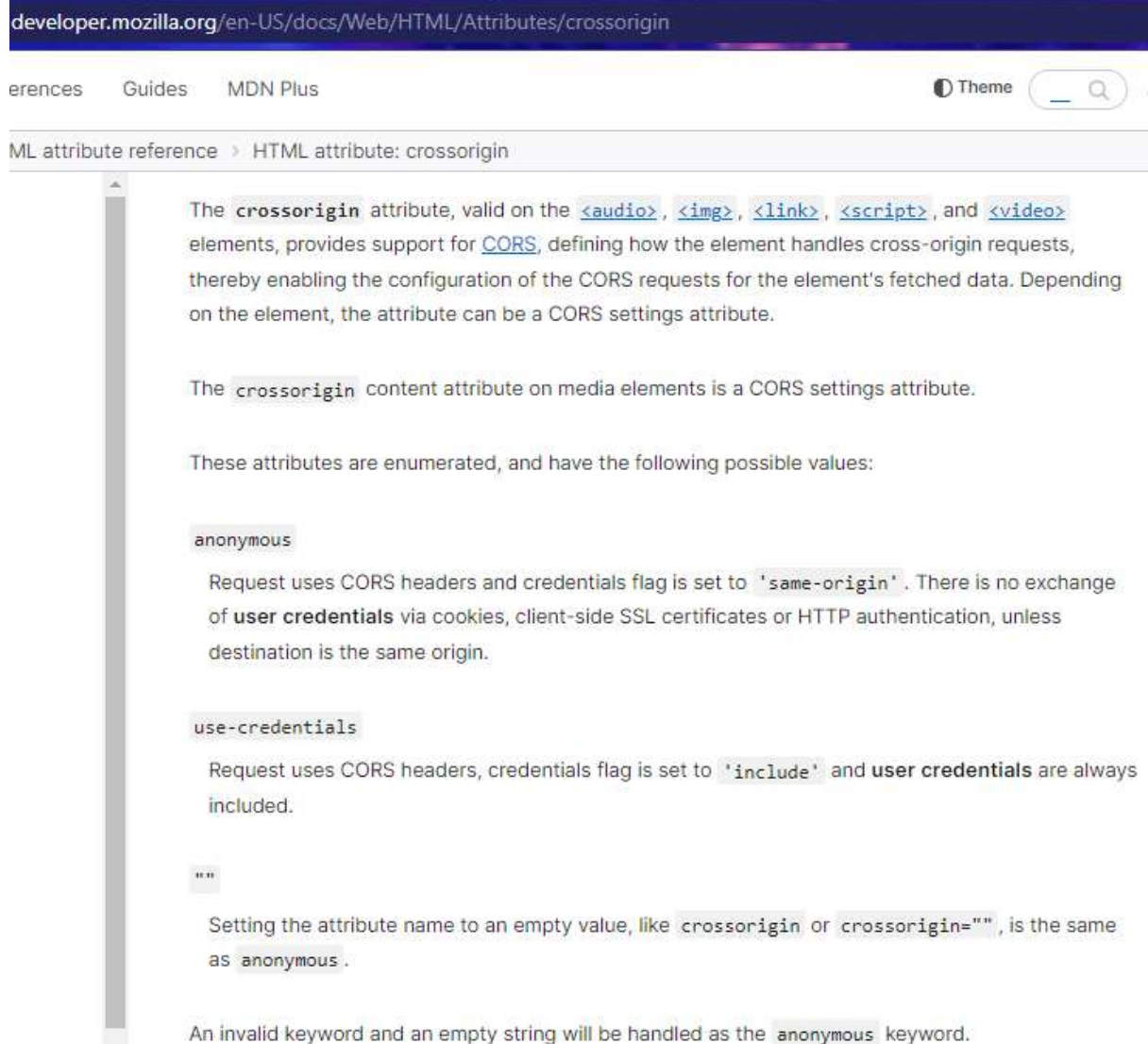
Integrity съдържа хеш стойност. Когато браузърът сваля .min. файла от линка на CDN, произчислява хеш стойността и я сравнява със стойността, която ние сме задали. Ако съвпадат,

знаем, че изтегляния файл не е бил променен. Ако не съвпадат, браузърът ще блокира изтеглянето.

При мен има още един нов атрибут:

```
integrity="sha384-gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5AR6csBvApHhNL/vI1Bx"  
crossorigin="anonymous">
```

CORS is **Cross Origin Resource Sharing**. It is a W3C standard that allows a server to make cross-domain calls from the specified domains, while rejecting others. By default, due to browser security, it prevents a web page from making one domain Ajax request to another domain.



The screenshot shows the MDN (Mozilla Developer Network) page for the HTML attribute `crossorigin`. The page title is "HTML attribute: crossorigin". The main content explains that the `crossorigin` attribute, valid on `<audio>`, `<img>`, `<link>`, `<script>`, and `<video>` elements, provides support for CORS, defining how the element handles cross-origin requests. It mentions that the attribute can be a CORS settings attribute. The page lists the possible values for the attribute: `anonymous`, `use-credentials`, and `""`. For `anonymous`, it states that the request uses CORS headers and the credentials flag is set to `'same-origin'`, with no exchange of user credentials. For `use-credentials`, it states that the request uses CORS headers, the credentials flag is set to `'include'`, and user credentials are always included. For `""`, it states that setting the attribute name to an empty value, like `crossorigin` or `crossorigin=""`, is the same as `anonymous`. It also notes that an invalid keyword and an empty string will be handled as the `anonymous` keyword.

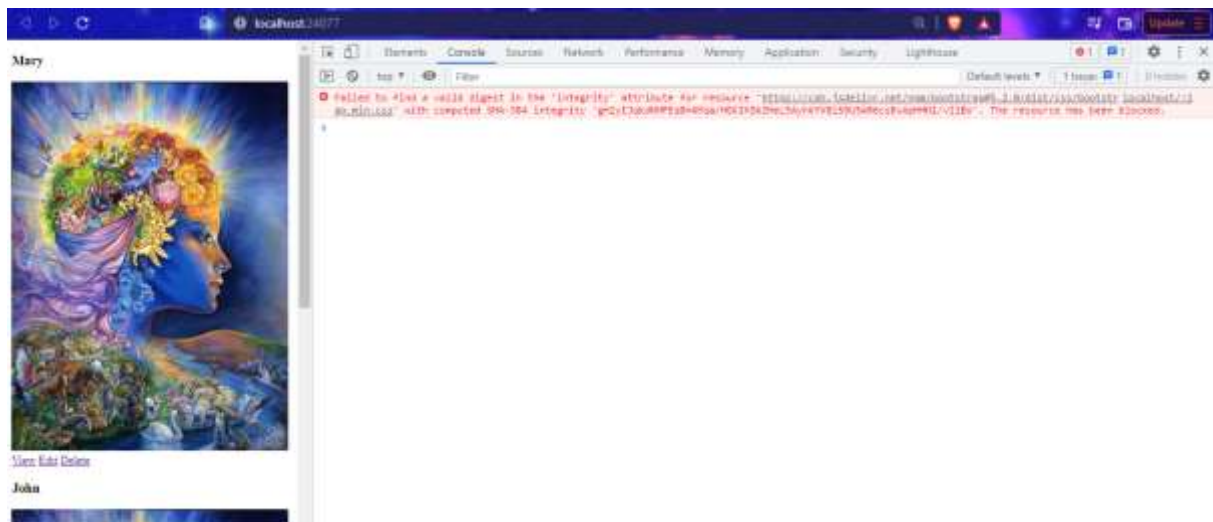
За да демонстрираме злонамерена намеса, ще променим последната стойност на integrity:

```
integrity="sha384-gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5AR6csBvApHhNL/vI1Bx"
```

\_wrong\_hash:

```
integrity="sha384-gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5_wrong_hash"
```

Refresh:



Bootstrap стиловете вече не работят, защото изтеглянето им от CDN сървъра е блокирано.

If the CDN:

- \* is down
- \* the app is not able to reach the CDN
- \* someone has maliciously altered the bootstrap file

the browser will fail to download the bootstrap file from CDN and the app will not work.

So we want in this case the app to go back to our own server and load the bootstrap.min from our server like the environment is Staging, Production, UAT....

Т.е. ако проверката между линк CDN-integrity не съвпада, приложението със стиловете не се зарежда в браузъра правилно и искаме да се върне и изтегли .min от нашия собствен сървър, както би направило, ако environment не е development.

## CDN and Fallback Source

```
<environment exclude="Development">
  <link rel="stylesheet"
    href="https://CDN_URL/css/bootstrap.min.css"
    integrity="sha384-Integrity_Hash...."
    our server asp-fallback-href="~/lib/bootstrap/css/bootstrap.min.css"
    asp-fallback-test-class="sr-only" sr=server
    asp-fallback-test-property="position"
    asp-fallback-test-value="absolute"
    asp-suppress-fallback-integrity="true">turn off checking
  </environment> checking we trust in our own server
```

➤ Use **asp-fallback-href** attribute to specify a fallback source

➤ Use **asp-suppress-fallback-integrity** attribute to turn off integrity check for files downloaded from fallback source

3 - test – helpers – тестват дали браузърът успешно е изтеглил bootstrap от CDN като инжектират custom JavaScript в браузъра, който проверява последователно за тези {–class-property–value} и ако те съществуват, т.е. създадени са, изтеглянето на bootstrap от CDN е успешно. Ако те не съществуват, т.е. браузърът не е изтеглил успешно bootstrap от CDN, изтегля .min от нашия собствен сървър, при което по подразбиране проверката е вярна, integrity = "true", защото няма вероятност от злонамерена намеса в нашия сървър. Нуждаем се от integrity проверка



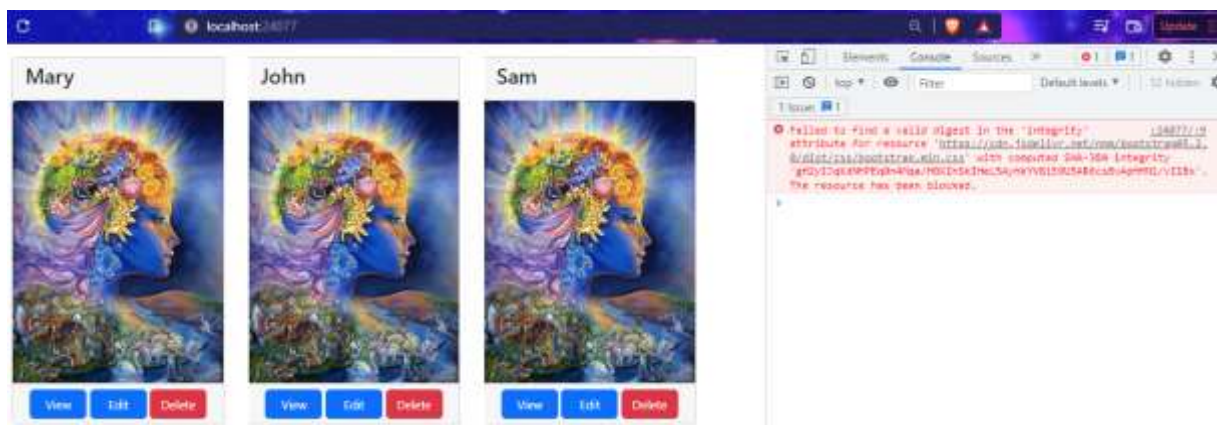
само за изтегляне от външния сървър на 3-и лица, както е CDN в случая. Ние имаме доверие на файловете, които се изтеглят от нашия сървър.

integrity = "true" – проверката е вярна за нашия сървър,

= "false" – или без integrity – (по подразбиране) = провери.

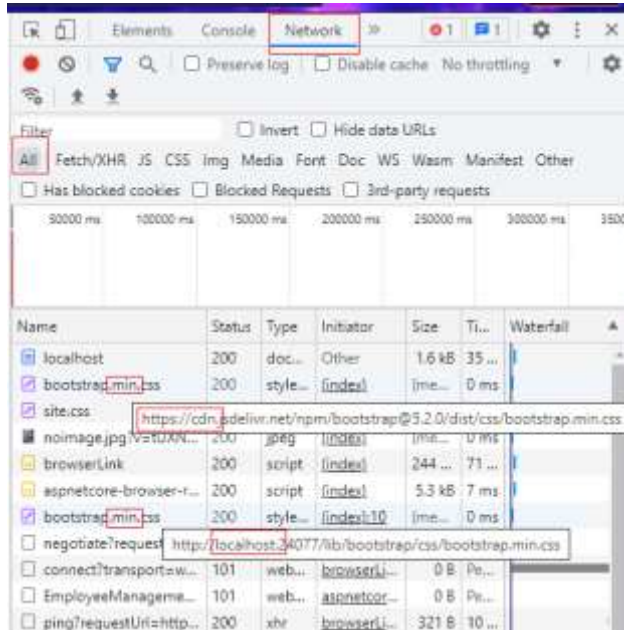
```
<environment include="Development">
  <link href="/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
</environment>
<environment exclude="Development">
  <!-- CSS only -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
    rel="stylesheet"
    integrity="sha384-gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5_wrong_hash"
    crossorigin="anonymous"
    asp-fallback-href="/lib/bootstrap/css/bootstrap.min.css"
    asp-fallback-test-class="sr-only"
    asp-fallback-test-property="position"
    asp-fallback-test-value="absolute"
    asp-suppress-fallback-integrity="true">
</environment>
```

```
<environment include="Development">
  <link href="/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
</environment>
<environment exclude="Development">
  <!-- CSS only -->
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
    rel="stylesheet"
    integrity="sha384-
gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkYV8i59U5_wrong_hash"
    crossorigin="anonymous"
    asp-fallback-href="/lib/bootstrap/css/bootstrap.min.css"
    asp-fallback-test-class="sr-only"
    asp-fallback-test-property="position"
    asp-fallback-test-value="absolute"
    asp-suppress-fallback-integrity="true">
</environment>
```

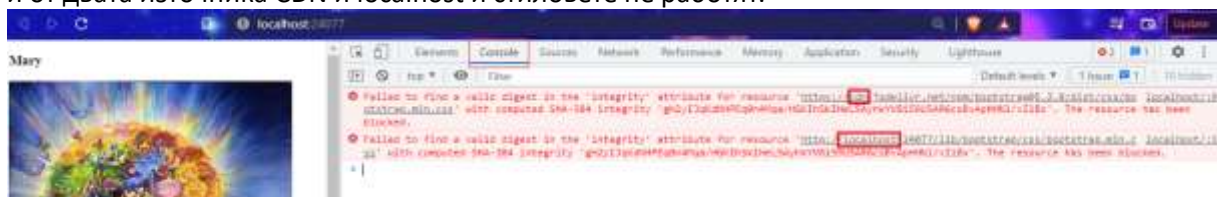


Изтеглянето е блокирано, защото хеш стойността не съвпада, затова е пренасочено към

изтегляне на .min от нашия сървър и стиловете работят.



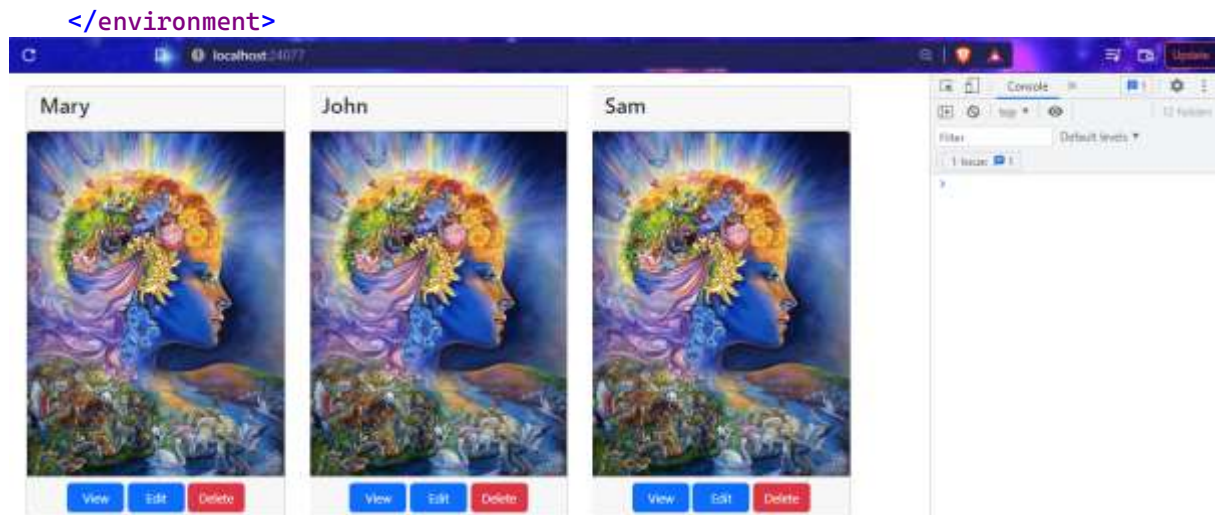
Ако зададем `asp-suppress-fallback-integrity="false">` ще извършва проверка дори при изтегляне на файла от нашия сървър и понеже сме задали `_wrong_hash`, изтеглянето се проваля и от двата източника CDN и localhost и стиловете не работят.



Коригираме всичко:

```
<environment include="Development">
  <link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
</environment>
<environment exclude="Development">
  <!-- CSS only -->
  <link
    href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
    rel="stylesheet"
    integrity="sha384-
gH2yIJqKdNHPEq0n4Mqa/HGKIhSkIHeL5AyhkyVV8i59U5AR6csBvApHhNL/vI1Bx"
    crossorigin="anonymous"
    asp-fallback-href="~/lib/bootstrap/css/bootstrap.min.css"
    asp-fallback-test-class="sr-only"
    asp-fallback-test-property="position"
    asp-fallback-test-value="absolute"
    asp-suppress-fallback-integrity="true">
```

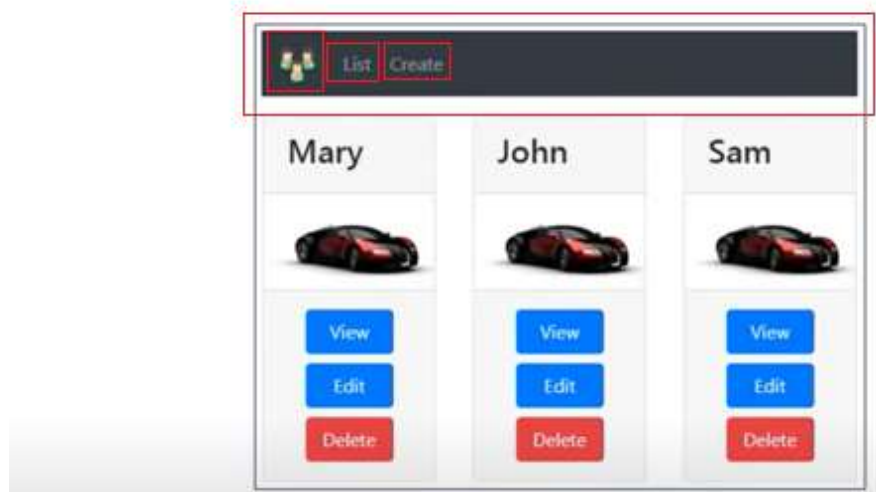




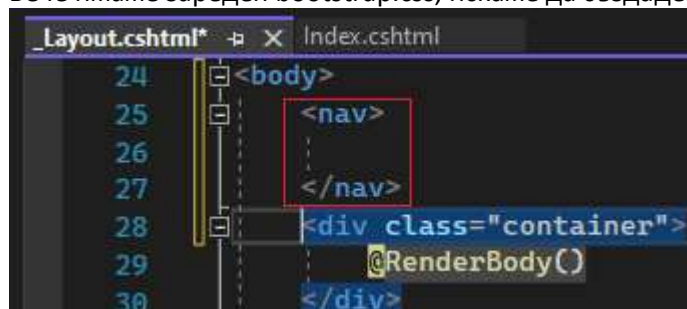
Работи добре.

launchsettings.json -> "ASPNETCORE\_ENVIRONMENT": "Development",

## 39 – Bootstrap Navigation Menu On a Large Screen Device



Вече имаме зареден bootstrap.css, искаме да създадем меню с bootstrap-navbar-class:



Първо `<nav>`-елемент, после в него влягаме bootstrap class.

`<nav>` - navigation menu

`<ul>` - unordered list

`<li>` - list item

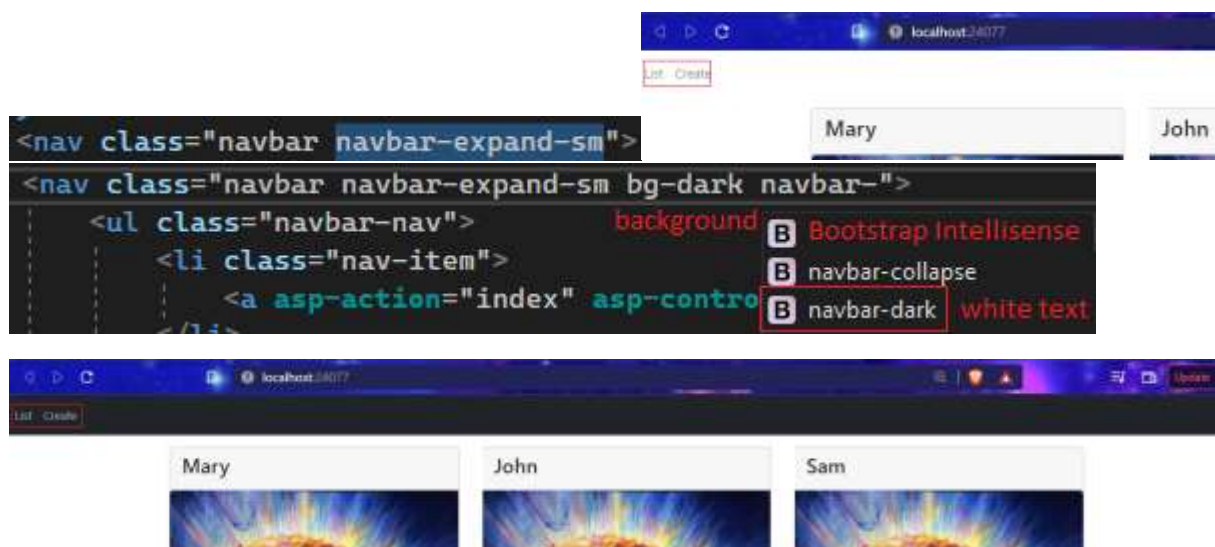
`<a>` - anchor

```
Layout.cshtml* X Index.cshtml
<body>
  <nav class="navbar">
    <ul class="navbar-nav">
      <li class="nav-item">
        <a asp-action="index" asp-controller="home" class="nav-link">List</a>
      </li>
      <li class="nav-item">
        <a asp-action="create" asp-controller="home" class="nav-link">Create</a>
      </li>
    </ul>
  </nav>
  <div class="container">
    @RenderBody()
  </div>
</body>
```

```
<nav class="navbar">
  <ul class="navbar-nav">
    <li class="nav-item">
      <a asp-action="index" asp-controller="home" class="nav-link">List</a>
    </li>
    <li class="nav-item">
      <a asp-action="create" asp-controller="home" class="nav-link">Create</a>
    </li>
  </ul>
</nav>
```

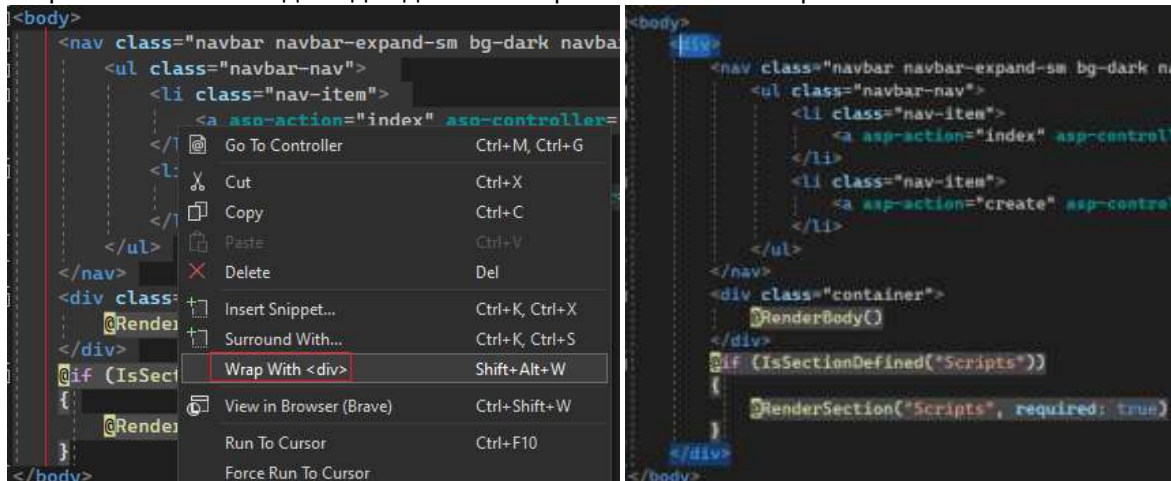


горизонтально меню: `<nav class="navbar navbar-expand-sm">`

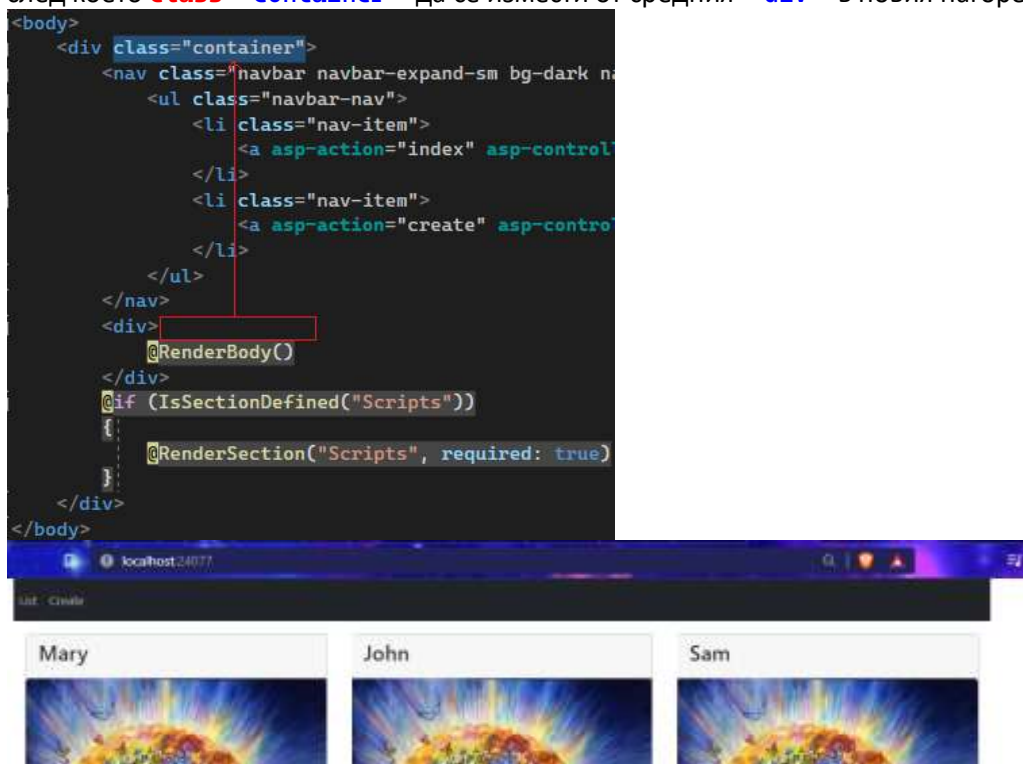


```
<nav class="navbar navbar-expand-sm">
  <nav class="navbar navbar-expand-sm bg-dark navbar->
    <ul class="navbar-nav">
      <li class="nav-item">
        <a asp-action="index" asp-controller="home" class="nav-link">List</a>
      </li>
      <li class="nav-item">
        <a asp-action="create" asp-controller="home" class="nav-link">Create</a>
      </li>
    </ul>
  </nav>
</nav>
```

Ширината на менюто да бъде еднаква с ширината на контейнера.

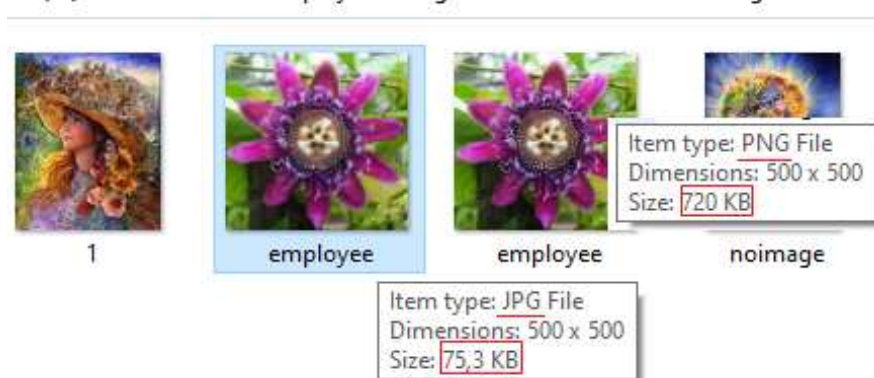


след което `class="container"` да се измести от средния `<div>` в новия нагоре.



Искаме и лого в навигационното меню:

иск (D:) > Exercises > EmployeeManagement > wwwroot > images

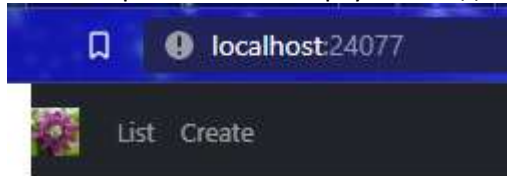


Избирам по-малката големина .JPG



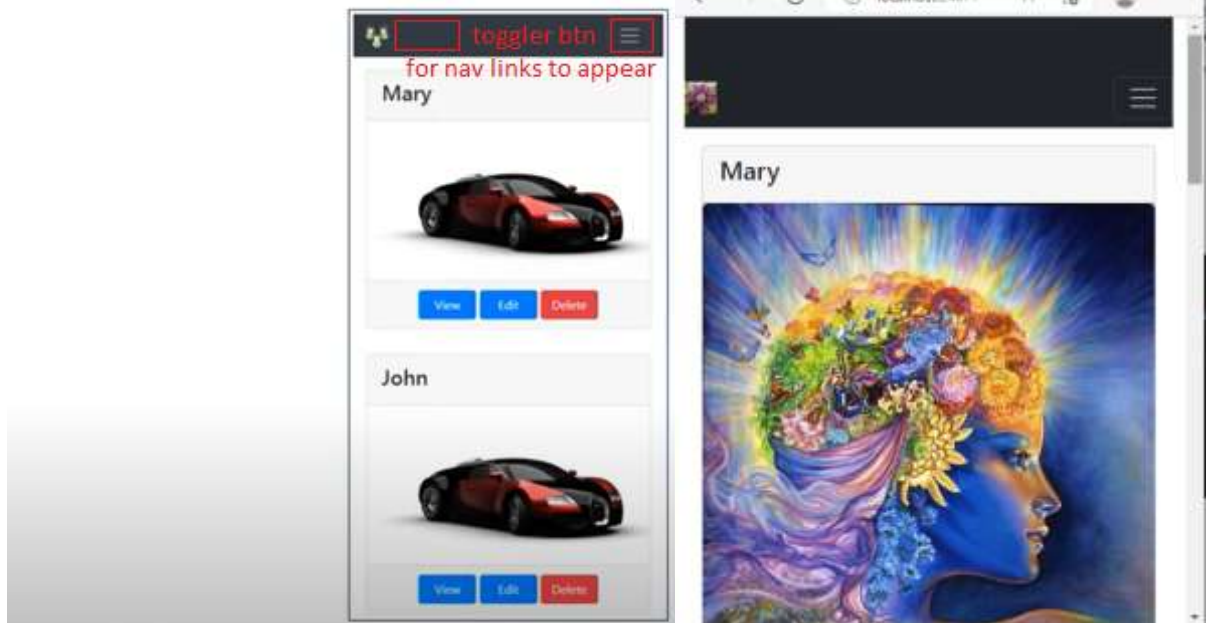
```
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <a class="navbar-brand" asp-action="index" asp-controller="home">
    
  </a>
</nav>
```

Искаме при натискане върху логото да ни препраща линк към началната страница.



```
<div class="container">
  <nav class="navbar navbar-expand-sm bg-dark navbar-dark">
    <a class="navbar-brand" asp-action="index" asp-controller="home">
      
    </a>
    <ul class="navbar-nav">
      <li class="nav-item">
        <a asp-action="index" asp-controller="home" class="nav-
link">List</a>
      </li>
      <li class="nav-item">
        <a asp-action="create" asp-controller="home" class="nav-
link">Create</a>
      </li>
    </ul>
  </nav>
  <div>
    @RenderBody()
  </div>
  @if (IsSectionDefined("Scripts"))
  {
    @RenderSection("Scripts", required: true)
  }
</div>
```

## `</div>` On a Small Screen Device



При мен работи само в Edge и пак стилирането не е много добре.

# = id selector.

```
<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <a class="navbar-brand" asp-action="index" asp-controller="home" >
    
  </a>
  <button type="button" class="navbar-toggler" data-toggle="collapse" data-target="#collapsibleNavbar">
    <span class="navbar-toggler-icon"></span>
  </button>
  <div class="collapse navbar-collapse" id="collapsibleNavbar">
    <ul class="navbar-nav">
      <li class="nav-item">
        <a asp-action="index" asp-controller="home" class="nav-link">List</a>
      </li>
      <li class="nav-item">
        <a asp-action="create" asp-controller="home" class="nav-link">Create</a>
      </li>
    </ul>
  </div>
</nav>
```

Тук се получи един много важен момент, който забелязвам и други пъти в рамките на VisualStudio – дори да има верен код, докато не се използва Intellisense, много пъти остава неразпознат и не работи, затова трябва да се обръща внимание дали цветовете са разпознати.

```
Details.cshtml _Layout.cshtml _ViewStart.cshtml _ViewStart.cshtml Test.cshtml
<script src="~/lib/bootstrap/js/bootstrap.esm.min.js"></script>
</environment>
<link href="~/css/site.css" rel="stylesheet" />
<title>@ViewBag.Title</title>
</head>
<body>
  <div class="container">
    <nav class="navbar navbar-expand-sm bg-dark navbar-dark">
      <a class="navbar-brand" asp-action="index" asp-controller="home" ></a>
      <a class="navbar-brand" asp-action="index" asp-controller="home" >
        
      </a>
```

```
</body>
  <div class="container">
```

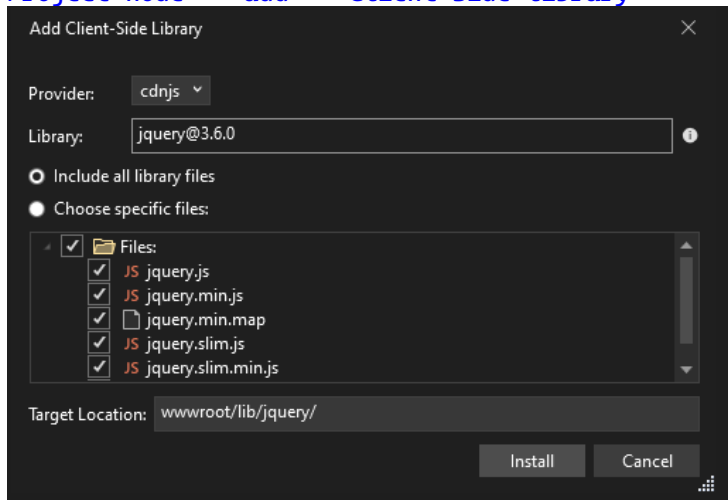
```

<nav class="navbar navbar-expand-sm bg-dark navbar-dark">
  <a class="navbar-brand" asp-action="index" asp-controller="home" >
    
  </a>
  <button type="button" class="navbar-toggler" data-toggle="collapse"
data-target="#collapsibleNavbar">
    <span class="navbar-toggler-icon"></span>
  </button>
  <div class="collapse navbar-collapse" id="collapsibleNavbar">
    <ul class="navbar-nav">
      <li class="nav-item">
        <a asp-action="index" asp-controller="home" class="nav-
link">List</a>
      </li>
      <li class="nav-item">
        <a asp-action="create" asp-controller="home" class="nav-
link">Create</a>
      </li>
    </ul>
  </div>
</nav>
<div>
  @RenderBody()
</div>
@if (IsSectionDefined("Scripts"))
{
  @RenderSection("Scripts", required: true)
}
</div>
</body>

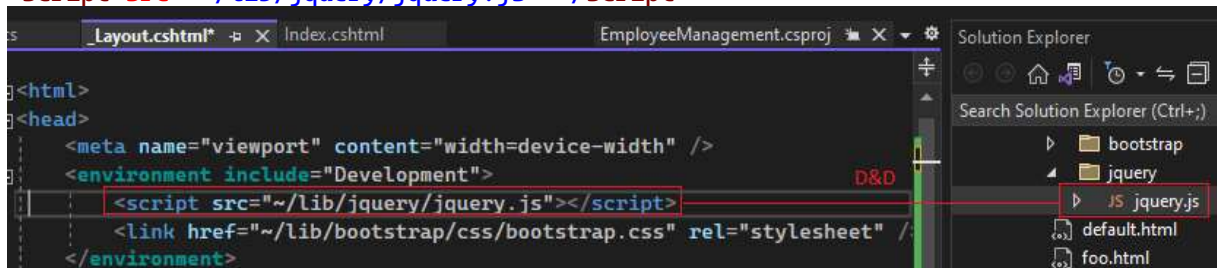
```

Toggler btn не работи, защото му е необходима JQuery функционалност.

Project node -> add -> Client side library



```
<script src="~/lib/jquery/jquery.js"></script>
```



Трябва ни също и Bootstrap JavaScript.



```

<meta name="viewport" content="width=device-width" />
<environment include="Development">
  <link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
  <script src="~/lib/jquery/jquery.js"></script>
  <script src="~/lib/bootstrap/js/bootstrap.js"></script>
</environment>
<environment exclude="Development">

```

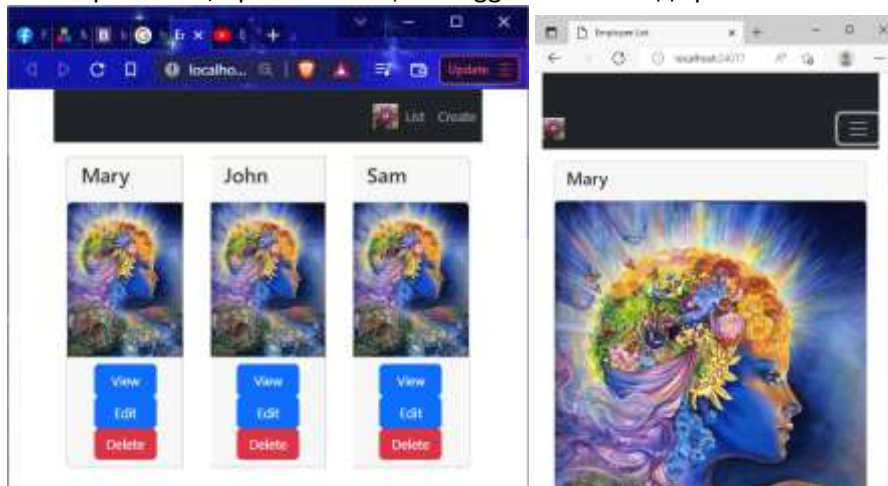
Последователността е много важна: JQuery трябва да бъде зареден преди bootstrap.js!

```

<link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
<script src="~/lib/jquery/jquery.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.js"></script>

```

Ако се размести, превключващият Toggler btn няма да работи.



При мен в Brave така и не се показва Toggler, но бутоните са подредени и искаме разстояния между тях. В Edge може да бъде селектиран Toggler, но не се разлиства.

Copy->Paste за двата вида Environment, съответно, единият е min.

```

<script src="~/lib/jquery/jquery.slim.min.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.esm.min.js"></script>

```

```

<environment include="Development">
  <link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
  <script src="~/lib/jquery/jquery.js"></script>
  <script src="~/lib/bootstrap/js/bootstrap.js"></script>
</environment>
<environment exclude="Development">
  <!-- CSS only -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.2.0/dist/css/bootstrap.min.css"
    rel="stylesheet"
    integrity="sha384-gH2yIJqKdNHPEq0n4Mqqa/HGKIhSkIHeL5AyhkYV8i59U5AR6csBvApHHNl/vI1Bx"
    crossorigin="anonymous"
    asp-fallback-href="~/lib/bootstrap/css/bootstrap.min.css"
    asp-fallback-test-class="sr-only"
    asp-fallback-test-property="position"
    asp-fallback-test-value="absolute"
    asp-suppress-fallback-integrity="true">
  <script src="~/lib/jquery/jquery.slim.min.js"></script>
  <script src="~/lib/bootstrap/js/bootstrap.esm.min.js"></script>
</environment>

```

```

<a asp-controller="home" asp-action="details" asp-route-id="@Employee.Id"
  class="btn btn-primary m-1">View</a>
<a href="#" class="btn btn-primary m-1">Edit</a>

```

```
<a href="#" class="btn btn-danger m-1">Delete</a>
```

```
Index.cshtml X margin-1
class="btn btn-primary m-1">View</a>
<a href="#" class="btn btn-primary m-1">Edit</a>
<a href="#" class="btn btn-danger m-1">Delete</a>
```



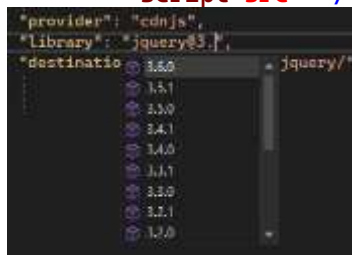
Margin m-1

Засега само List работи от менюто, следващият път ще добавим Create action in HomeController + Create View.

Brave, Mozilla, Chrome показват по еднакъв начин сайта, Edge по-различен.

При мен Toggler btn не работи дори с инсталация на по-стара версия на JQuery 3.3.1 като в туториъла и при правилна подредба:

```
<link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
<script src="~/lib/jquery/jquery.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.js"></script>
```



```
<link href="~/lib/bootstrap/css/bootstrap.css" rel="stylesheet" />
<script src="~/lib/jquery/jquery.js"></script>
<script src="~/lib/bootstrap/js/bootstrap.js"></script>
```

Впоследствие откраднах от другия туториъл C# crash course автокода за toggler:

```
<body>
<div class="container-fluid">
<nav class="navbar navbar-expand-sm navbar-toggleable-sm bg-dark navbar-dark border-bottom box-shadow mb-3">
<a class="navbar-brand" asp-area="" asp-controller="home" asp-action="index">

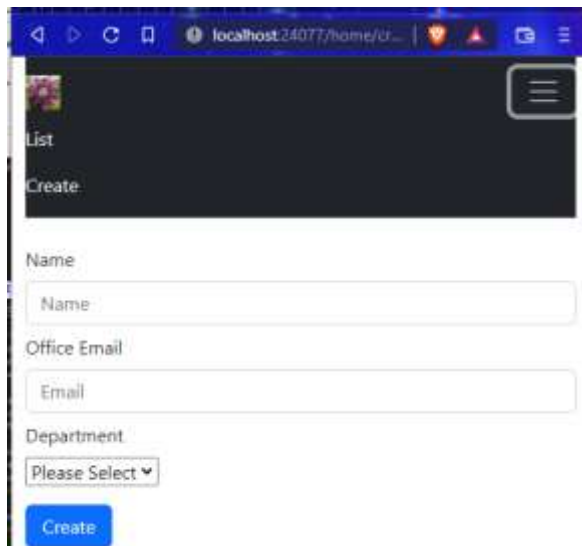
</a>
<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target=".navbar-collapse"
aria-controls="navbarSupportedContent" aria-expanded="false" aria-label="Toggle navigation">
<span class="navbar-toggler-icon"></span>
</button>
<div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
<ul class="navbar-nav flex-grow-1">
<li class="nav-item">
<a class="nav-link text-light" asp-area="" asp-controller="home" asp-action="index">List</a>
</li>
<li class="nav-item">
<a class="nav-link text-light" asp-area="" asp-controller="home" asp-action="create">Create</a>
</li>
</ul>
</div>
</div>
</nav>
<div>
RenderBody()
</div>
```

```
<div class="container-fluid">
<nav class="navbar navbar-expand-sm navbar-toggleable-sm bg-dark navbar-dark
border-bottom box-shadow mb-3">
```

```

<a class="navbar-brand" asp-area="" asp-controller="home" asp-action="index">
    
</a>
<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target=".navbar-collapse" aria-controls="navbarSupportedContent" aria-
expanded="false" aria-label="Toggle navigation">
    <span class="navbar-toggler-icon"></span>
</button>
<div class="navbar-collapse collapse d-sm-inline-flex justify-content-between">
    <ul class="navbar-nav flex-grow-1">
        <li class="nav-item">
<a class="nav-link text-light" asp-area="" asp-controller="home" asp-
action="index">List</a>
        </li>
        <li class="nav-item">
<a class="nav-link text-light" asp-area="" asp-controller="home" asp-
action="create">Create</a>
        </li>
    </ul>
</div>
</nav>

```



Name

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Department  
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